

Cheng Dai

School of Science, Westlake University

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EDUCATION

Westlake University, WLU

Sep. 2025 - Jul. 2026

Research Assistant

Supervisors: [Xin Yuan](#) and [Fanglin Bao](#)

- Dissertation Fields: HADAR (nature2023, Temperature-emissivity-texture-distance decomposition)
- AI department manager in the flab. Conduct Research and guide students.
- **Release the first thermal hyperspectral image restoration methods based on Radiative Transfer Equation, HAIR.**
- **Release the world first temperauture-emissivity-texture decomposition dataset and benchmark, TeX-1500.**

Wuhan University, WHU

Sep. 2025 - Mar. 2026

Academic Master's Degree Student (Withdrawal) in State Key Laboratory of Surveying and Mapping Remote Sensing Information Engineering, Major in Communication Engineering

Supervisors: [Wei He](#)

- GPA: 90.79%. First-class entrance scholarship.

Jilin University, JLU

Sep. 2021 - Jun. 2025

BSc. Major in Communication Engineering

- GPA: 90.58%. Rank: 12/283 (Outstanding graduate, top 5%).
- Outstanding Undergraduate Thesis (top 3%).
- Outstanding Student of the School (*4). First-Class Scholarship, Second-Class Scholarship (*3)
- 508/710 in CET6. Intermediate Russian excellent. One-year German Study.

PUBLICATIONS

[1] **Cheng Dai**, Jiale Lin, Hongyi Xu, Bingxuan Song, Ziyang Xie, Fanglin Bao

TeX-1500: A Paired Real-World LWIR Hyperspectral Dataset and Benchmark for Temperature-Emissivity-Texture Decomposition. [[Paper](#)], [[Homepage](#)], [[huggingface](#)] (IEEE Transactions on Image Processing under review, IF=13.7)

[2] **Cheng Dai**, Jiale Lin, Bingxuan Song, Yifei Chen, Jiashuo Chen, Xin Yuan, and Fanglin Bao

HAIR: HADAR-Based Thermal Infrared Hyperspectral Image Restoration. [[Paper](#)], [[Homepage](#)] (arxiv version)

INTERN EXPERIENCE

Changchun Institute of Optics, CAS, Science and Technology Innovation Plan

Aug. 2023 - Sep. 2024

- As a student leader, collect infrared military videos, and train small target detection via Fast-RCNN, DETR, EfficientDNet .etc. Finally end the project backbone via DilatedUNet, have the award “Excellent Completion”.

Lightwheel, Embodied Assets Group

Dec. 2024 - Feb. 2025

Group Leader: [Jian Yang](#), and [Soroush Nasiriany](#)

- Built tools for embodied simulation, asset construction, data collection, and teleoperation with Blender, MuJoCo, Robosuite, Issacsim and RoboCasa. [[Few Contribute Codes](#), Merged in Final [Robocasa](#)]
- Achievement: develop spacemouse, blender2yaml, yaml2blender plugins, reducing validations cycles 20× Faster.

Amoon AI, Large Audio Model Department

Apr. 2025 - Sep. 2025

- Independently collect and label 20k infant-cry audios (update 6 versions), and train or finetune models to classify the baby cries and detect activities. Involved Models: BEATs, Audio-MAE, Wav2vec2, Qwen-audio, Whisper, Conformer.

- Achievements: reach 67.2% acc in five-class (diaper, hunger, hug, sleepy, gasping) edge deployment device.

COMPETITION AWARDS

- First Prize in the Provincial Level of the National Undergraduate Mathematics Competition.
- Honorable Mention in the American Mathematical Competition.

SKILLS

Fields (from greatest strength to least strength):

- **Physical Inversion, Numerical Computation, Hyperspectral Image Processing.**
- **Improvements to Generative Model (Multi-branch diffusion, Schrodinger Bridge .etc.) and Interpretability Analysis (i.e. Fisher Information theory, Convex Theory, Derivation of differential equations).**
- **Train and Finetune Generative Model, Large Language Model, Vision Language Model, Large Audio Model.**
- **Model-based and deep-learning-based speech signal processing, image denoising algorithms.**

Software: [Cuda-Jax, Python, MATLAB, LaTeX, C++, Html5, CSS3, Javascript, Codex].

PERSONAL WORDS

- **I can do research assistant at anytime and any place on-site at anytime.**
- Never commit academic fraud! Keep working honestly!
- I'm hardworking. I will work 100 hours a week in areas that interest me.
- Beg for an opportunity for phd or highly-paid research assistant.